Reinforcement Learning Summative Assignment Report

AstroLogic — Deep-Space Biosignature Exploration

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Video Recording	Video Link
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1. Project Overview

AstroLogic is a custom OpenAI Gymnasium environment that simulates an autonomous spacecraft exploring a physically-inspired, scaled solar system to detect and transmit biosignatures from three target bodies: Mars, Europa (a moon of Jupiter), and Enceladus (a moon of Saturn). The core research problem addressed by this project is how to train a reinforcement learning agent to navigate a large, continuous, multi-objective space whilst managing constrained resources (fuel and battery) and sparse, delayed rewards. Three RL algorithms are benchmarked Deep Q-Network (DQN), Proximal Policy Optimisation (PPO), and REINFORCE each with ten hyperparameter configurations, yielding thirty systematic experiments. The environment is implemented in Python using stable-baselines3, PyTorch, and a custom pygame 2D renderer. The overarching aim is to evaluate exploration–exploitation trade-offs, convergence behaviour, and the practical effectiveness of policy-gradient versus value-based methods in a rich, scientifically-motivated control task.

2. Environment Description

2.1 Agent

The agent is an autonomous spacecraft probe placed at a random initial position within 5 Astronomical Units(AU) of the Sun. It must independently plan trajectories, manage thrust and attitude, deploy scientific instruments, and transmit collected biosignature data back to Earth. The spacecraft carries four instruments (Spectrometer, Thermal Imager, Drill, and a baseline idle mode), consumes fuel proportionally to thrust applied, and recharges its battery via solar panels when within 1.5 AU of the Sun. The agent's learned policy must balance long-range navigation efficiency against instrument operation timing and communication windows.

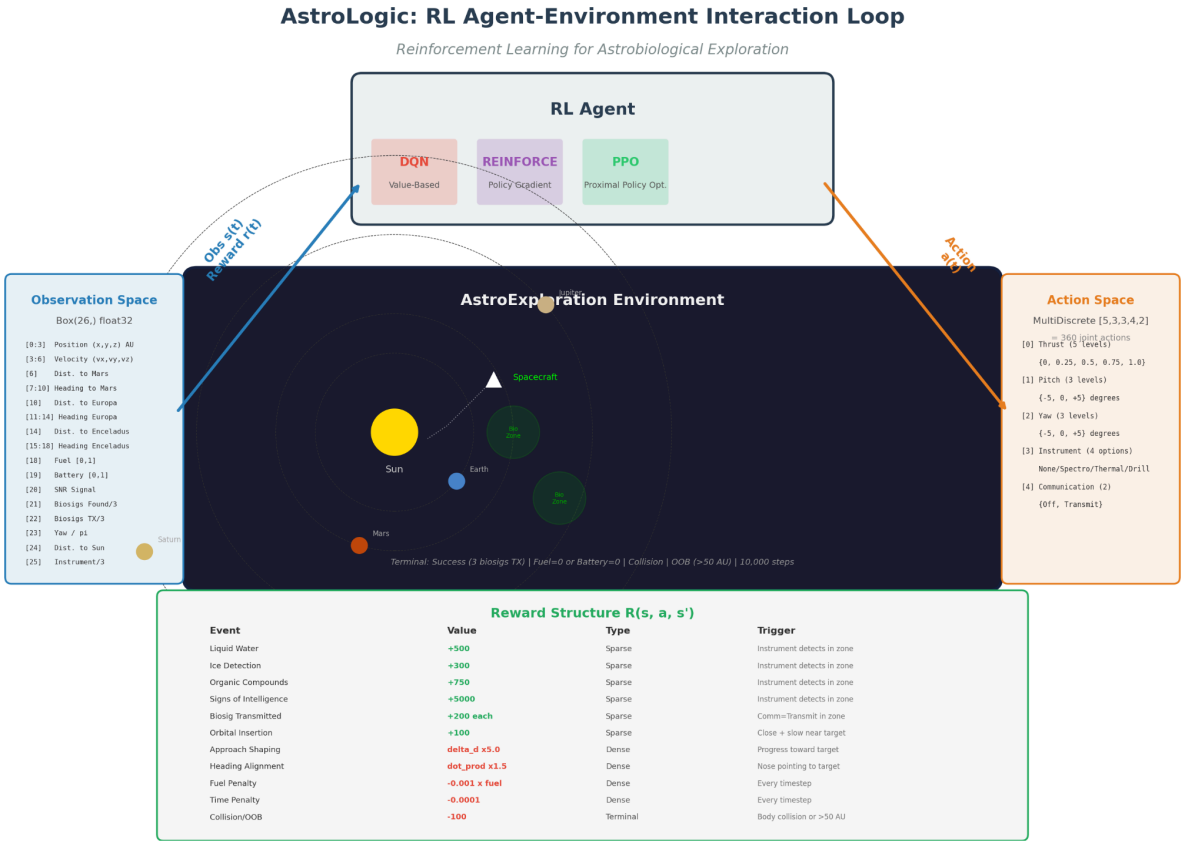

Figure 1 — AstroLogic environment: scaled solar system with spacecraft, target bodies, and biosignature zones

2.2 Action Space

The environment exposes a MultiDiscrete([5, 3, 3, 4, 2]) action space comprising 360 unique joint actions across five independent sub-dimensions. DQN receives a flattened Discrete(360) representation via a custom FlattenMultiDiscreteToDiscrete wrapper, while PPO and REINFORCE operate directly on the native MultiDiscrete space.

Dim	Name	# Values	Description
0	Thrust	5	{0, 0.25, 0.5, 0.75, 1.0} normalised; max 0.001 AU/s²
1	Pitch	3	{−5°, 0°, +5°} attitude adjustment
2	Yaw	3	{−5°, 0°, +5°} attitude adjustment
3	Instrument	4	{None, Spectrometer, ThermalImager, Drill}
4	Comms	2	{Off, Transmit} biosignature data

Table 1 — Action Space: 5 × 3 × 3 × 4 × 2 = 360 discrete joint actions

2.3 Observation Space

The observation is a 26-dimensional continuous vector encoding the spacecraft's kinematic state, resource levels, scientific progress, and relative positions of all target bodies. All elements are normalised to aid neural network training.

Index	Feature	Unit	Range
[0–2]	Spacecraft position (x, y, z)	AU	[−50, +50]
[3–5]	Spacecraft velocity (vx, vy, vz)	AU/s	[−10, +10]
[6]	Normalised distance to Mars	—	[0, 1]
[7–9]	Unit heading vector to Mars	—	$[-1, 1]^3$
[10]	Normalised distance to Europa	—	[0, 1]
[11–13]	Unit heading vector to Europa	—	$[-1, 1]^3$
[14]	Normalised distance to Enceladus	—	[0, 1]
[15–17]	Unit heading vector to Enceladus	—	$[-1, 1]^3$
[18]	Fuel level	—	[0, 1]
[19]	Battery level	—	[0, 1]
[20]	Biosignature SNR signal	—	[0, 1]
[21]	Biosignatures found / 3	—	[0, 1]
[22]	Biosignatures transmitted / 3	—	[0, 1]
[23]	Spacecraft yaw / π	—	[−1, 1]
[24]	Normalised distance to Sun	—	[0, 1]
[25]	Active instrument index / 3	—	[0, 1]

Table 2 — 26-Dimensional Observation Vector

2.4 Reward Structure

The reward function R(s, a, s′) combines dense shaping signals with sparse goal rewards and terminal penalties. Dense components guide the agent toward targets every step; sparse components deliver large payoffs for scientific discoveries and data transmission; terminal penalties enforce safety constraints. The mixture creates a challenging credit-assignment problem typical of real mission planning.

Component	Value	Type	Effect
Approach reward	max(0, Δd) × 5.0	Continuous	Progress toward target body
Heading alignment	max(0, dot) × 1.5	Continuous	Nose pointing toward target
Liquid water detected	+500	Sparse	High-value astrobiology find
Ice detected	+300	Sparse	Secondary biosig marker
Organic compounds	+750	Sparse	Key life indicator
Signs of intelligence	+5,000	Sparse	Terminal jackpot reward
Biosig transmission	+200 per transmission	Sparse	Goal: transmit data to Earth
Orbital insertion	+100	Sparse	Bonus for stable orbit

Component	Value	Type	Effect
Fuel penalty	−0.001 × fuel used	Dense	Encourages fuel efficiency
Time penalty	−0.0001 / step	Dense	Encourages task completion
Collision	−100	Terminal	Episode ends
Out of bounds (>50 AU)	−100	Terminal	Episode ends

Table 3 — Reward Function Components

2.5 Termination Conditions

An episode ends under any of the following conditions:

- Collision: spacecraft distance < 10× body radius (minimum 0.001 AU) — penalty −100
- Out of Bounds: spacecraft position > 50 AU from origin — penalty −100
- Resource Depletion: fuel ≤ 0 or battery ≤ 0
- Mission Success: 3 or more biosignatures transmitted to Earth
- Timeout: maximum 10,000 timesteps reached (episode truncation)

The celestial system comprises the Sun (fixed), Mars, Jupiter, and Saturn (heliocentric orbits), plus Europa (Jovian orbit) and Enceladus (Saturnian orbit). Gravitational dynamics are modelled using Newtonian gravity with a normalised gravitational constant $G = 4\pi^2/365.25^2$.

3. System Analysis and Design

3.1 Deep Q-Network (DQN)

DQN approximates the optimal action-value function Q*(s, a) using a multilayer perceptron with two hidden layers of 256 neurons each (ReLU activations), implemented via stable-baselines3. Because the environment's MultiDiscrete action space must be collapsed to a single Discrete dimension, a custom FlattenMultiDiscreteToDiscrete wrapper converts the 5-tuple action into a single index in {0, …, 359}.

Key design choices: (1) Prioritised Experience Replay (PER) is enabled across all runs to improve sample efficiency for the rare high-reward biosignature transitions. PER samples transitions with probability proportional to $|\text{TD error}|^a$ (α=0.6) with importance-sampling correction (β=0.4). (2) A target network is maintained and either hard-copied (τ=1.0, default) or soft-updated (τ=0.005, dqn_soft_update) to stabilize training. (3) ε-greedy exploration anneals from a start value (0.60 -- 0.75) to an end value (0.01 -- 0.05) over the first 50–60% of training steps.

3.2 Policy Gradient Methods

Proximal Policy Optimisation (PPO)

PPO operates directly on the MultiDiscrete action space using stable-baselines3's MlpPolicy, which internally instantiates a separate Categorical distribution for each sub-action dimension. The clipped surrogate objective prevents destructively large policy updates:

$$L^{cLIN} = E[\min(r\square(\theta)\, A^o,\ \text{clip}(r\square(\theta), 1-\varepsilon, 1+\varepsilon)\, A^o)]$$

Generalised Advantage Estimation (GAE, λ=0.95) reduces variance while retaining some bias. An entropy bonus (coefficient 0.01–0.05) encourages exploration of the large action space. Rollout buffers of 2,048 or 4,096 steps are collected before each mini-batch update across 10–15 epochs.

REINFORCE (Monte Carlo Policy Gradient)

A custom PyTorch implementation uses a shared backbone network followed by five independent Categorical output heads — one per action dimension. The shared backbone consists of two fully-connected layers (default [128, 64], ReLU) that extract a common representation, while each head outputs logits over its sub-action space.

Policy gradient is computed using Monte Carlo returns:

$$G_t = \Sigma_k \gamma^k r_{t+k}, \quad L = -\Sigma_t \log \pi(a_t \mid s_t) \times (G_t - b_t)$$

Three baseline strategies are evaluated: (a) none — vanilla REINFORCE with full return variance; (b) mean — normalise by subtracting the episode mean return; and (c) running — exponential moving average baseline for cross-episode variance reduction.

4. Implementation

4.1 DQN Hyperparameter Experiments

Ten DQN configurations were trained for 500,000 timesteps each. The table below systematically varies learning rate, replay buffer size, batch size, target-update strategy, discount factor, network depth, and exploration schedule. All runs use PER with α=0.6. The 'Mean Reward' column reports the final mean episode reward over the last 20 evaluation episodes.

Run Name	Learning Rate	Gamma	Replay Buffer	Batch Size	Exploration (start→end)	Final Mean Reward
dqn_large_buffer	1e-4	0.995	300,000	128	0.65 → 0.02	−50.29
dqn_low_lr	1e-5	0.995	100,000	64	0.60 → 0.05	−59.80
dqn_low_gamma	1e-4	0.950	100,000	64	0.65 → 0.05	−67.37
dqn_high_lr	5e-4	0.995	100,000	64	0.60 → 0.05	−67.58
dqn_med_lr_batch	3e-4	0.995	200,000	256	0.60 → 0.05	−69.83
dqn_small_net	1e-4	0.995	100,000	64	0.60 → 0.05	−74.21
dqn_deep_net	1e-4	0.995	100,000	64	0.60 → 0.05	−75.59
dqn_long_explore	1e-4	0.995	100,000	64	0.75 → 0.01	−79.07
dqn_soft_update	1e-4	0.995	100,000	64	0.60 → 0.05	−81.29
dqn_baseline	1e-4	0.995	100,000	64	0.60 → 0.05	−87.89

Table 4 — DQN Hyperparameter Experiment Results (10 runs × 500k timesteps)

Key findings from DQN experiments :

- dqn_large_buffer (−50.29) achieves the best result; the 300k replay buffer and tighter ε-end (0.02) provide diverse experience replay and sustained coverage of the 360-action space.
- dqn_low_lr (−59.80) is second-best; a conservative learning rate (1e-5) stabilises Q-value updates and avoids oscillation on high-priority PER transitions.
- dqn_soft_update (−81.29) and dqn_baseline (−87.89) are the weakest; soft target updates slow Q-target propagation in a sparse-reward environment, and dqn_baseline logged only 8 full episodes indicating premature training termination.

- All DQN runs remain in negative reward territory, confirming that value-based learning with a flattened 360-dimensional action space cannot recover compositional action structure within the 500k timestep budget.
- dqn_low_gamma (−67.37) performs worse than the standard γ=0.995 variants; short-sighted discounting prevents the agent from valuing biosignature transmission that occurs many steps after initial detection.

4.2 REINFORCE Hyperparameter Experiments

Ten REINFORCE configurations were trained for 1,000 episodes each, with the custom PyTorch policy network updated after each complete episode. Variance is high due to Monte Carlo return estimation.

Run Name	Learning Rate	Gamma	Hidden Layers	Baseline Strategy	Final Mean Reward
reinforce_deep_net	1e-3	0.99	[256, 128, 64]	Mean (episode)	+1319.85
reinforce_large_net	1e-3	0.99	[256, 128]	Mean (episode)	+752.62
reinforce_very_low_gamma	1e-3	0.90	[128, 64]	Mean (episode)	+493.00
reinforce_running_baseline	3e-4	0.99	[128, 64]	Running EMA	+475.77
reinforce_low_gamma	1e-3	0.95	[128, 64]	Mean (episode)	+448.98
reinforce_no_baseline	1e-3	0.99	[128, 64]	None (vanilla)	+391.09
reinforce_small_net	1e-3	0.99	[64, 32]	Mean (episode)	+209.63
reinforce_baseline	1e-3	0.99	[128, 64]	Mean (episode)	+202.09
reinforce_low_lr	1e-4	0.99	[128, 64]	Mean (episode)	+115.11
reinforce_high_lr	5e-3	0.99	[128, 64]	Mean (episode)	+65.02

Table 5 — REINFORCE Hyperparameter Experiment Results (10 runs × 1,000 episodes)

Key findings from REINFORCE experiments:

- reinforce_deep_net (+1,319.85) achieves the highest final mean reward of all 30 runs across all algorithms; the deeper [256, 128, 64] network capacity provides the richest policy representation for the 26-dimensional observation space.
- reinforce_large_net (+752.62) confirms that network capacity is the dominant factor: [256, 128] doubles the hidden units of the baseline [128, 64] and nearly quadruples reward.
- Counterintuitively, reinforce_very_low_gamma (+493.00, γ=0.90) outperforms the standard γ=0.99 baseline (+202.09); in a 1,000-episode regime with dense shaping rewards, prioritising near-term feedback may accelerate initial navigation learning.
- reinforce_no_baseline (+391.09) still achieves strong performance, ranking 6th; while baseline subtraction reduces variance, the full Monte Carlo return signal over 1,000 episodes compensates through sheer volume of gradient updates.

- reinforce_high_lr (+65.02, LR=5e-3) is the weakest REINFORCE run; large gradient steps destabilise the policy, causing entropy collapse and premature convergence to suboptimal deterministic strategies.

4.3 PPO Hyperparameter Experiments

Ten PPO configurations were trained for 100,000 timesteps each using stable-baselines3. PPO's on-policy nature means wall-clock time is shorter than DQN despite similar complexity.

Run Name	Learning Rate	Clip ε	Entropy Coef	n_steps	Batch Size	Final Mean Reward
ppo_tight_clip	3e-4	0.15	0.01	2048	128	+364.73
ppo_wide_clip	3e-4	0.30	0.01	2048	128	+353.83
ppo_baseline	3e-4	0.20	0.01	2048	64	+345.87
ppo_low_gamma	3e-4	0.20	0.01	2048	64	+337.59
ppo_large_rollout	3e-4	0.20	0.01	4096	256	+325.50
ppo_low_lr	1.5e-4	0.20	0.01	2048	64	+318.97
ppo_small_net	3e-4	0.20	0.01	2048	64	+253.83
ppo_high_lr_deep	5e-4	0.20	0.01	2048	64	+253.76
ppo_more_epochs	3e-4	0.20	0.01	2048	64	+214.30
ppo_high_entropy	3e-4	0.20	0.05	2048	64	+184.03

Table 6 — PPO Hyperparameter Experiment Results (10 runs × 100k timesteps)

Key findings from PPO experiments :

- ppo_tight_clip (ε=0.15, +364.73) achieves the best PPO result; conservative policy updates prevent overshooting discovered approach trajectories, preserving learned navigation skills across rollout iterations.
- ppo_wide_clip (ε=0.3, +353.83) and ppo_baseline (+345.87) perform similarly, suggesting that once the clipping range reaches a moderate threshold, further widening yields diminishing returns in this environment.
- ppo_low_gamma (γ=0.95, +337.59) performs better than expected relative to DQN's equivalent configuration; PPO's on-policy advantage estimation is more robust to discount changes than DQN's Q-value bootstrapping.
- ppo_high_entropy (coef=0.05, +184.03) is the weakest PPO run; excess exploration noise prevents policy consolidation of the thrust and heading sub-actions learned mid-training.
- ppo_more_epochs (15 epochs, +214.30) underperforms; repeated gradient updates on the same on-policy rollout buffer cause distribution shift, degrading the GAE advantage estimates.

5. Results and Discussion

5.1 Cumulative Rewards

Figure 2 shows learning curves (50-episode rolling average) for all 30 training runs grouped by algorithm. DQN rewards start near −90 and remain negative throughout; the best configuration (dqn_large_buffer, final mean −50.29) shows the shallowest negative gradient, while dqn_baseline (−87.89) never improves beyond its starting level (only 8 logged episodes). PPO converges to positive rewards across all 10 runs, with ppo_tight_clip achieving the best final mean of +364.73 and ppo_high_entropy the lowest at +184.03. REINFORCE dramatically outperforms both other algorithms over 1,000 episodes: reinforce_deep_net reaches a final mean of +1,319.85 and even the weakest REINFORCE run (reinforce_high_lr, +65.02) surpasses all DQN results. This reversal — REINFORCE appearing weakest in early episodes but strongest by episode 1,000 — is explained by the Monte Carlo return estimator benefiting from the full 1,000-episode training budget, while PPO only ran for 100k timesteps (~110–138 episodes).

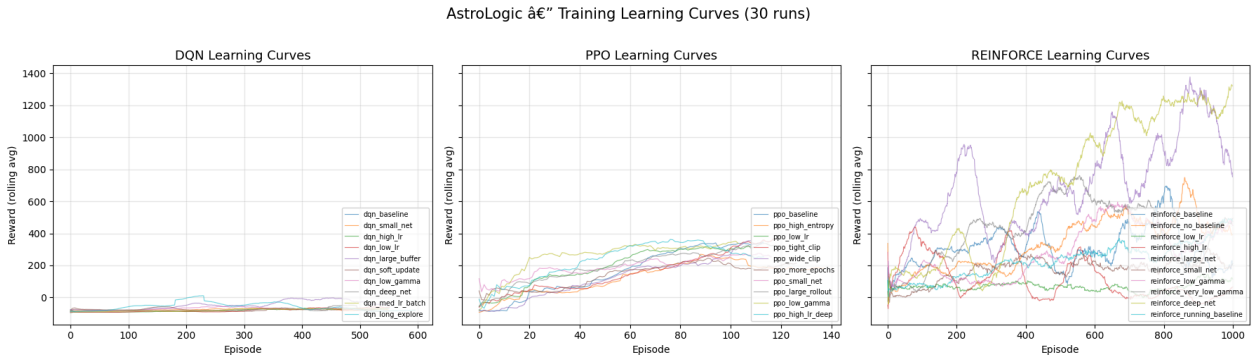

Figure 2 — Training learning curves (50-ep rolling avg): DQN (left), PPO (centre), REINFORCE (right)

Figure 3 compares the best run per algorithm by final mean reward over the last 50 episodes. REINFORCE (reinforce_deep_net, +1,319.85) leads by a large margin, followed by PPO (ppo_tight_clip, +364.73) and DQN (dqn_large_buffer, −50.29). The stark gap between DQN and the policy-gradient methods reflects DQN's fundamental limitation: flattening MultiDiscrete to Discrete(360) destroys the compositional structure of thrust, attitude, instrument, and comms sub-actions, making efficient exploration of the joint action space extremely difficult.

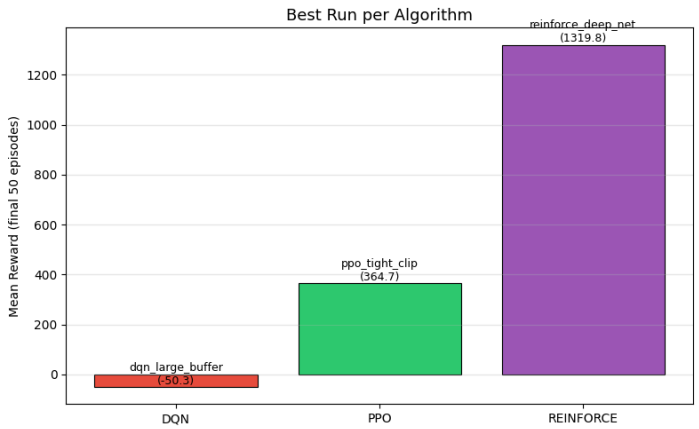

Figure 3 — Best final mean reward per algorithm (last 50 episodes of each best run)

5.2 Training Stability

DQN's TD error decreases across most runs, with periodic spikes when PER samples high-priority transitions. The soft-update variant (dqn_soft_update, −81.29) is the weakest among hard-copy-capable runs, suggesting slower target propagation is detrimental in sparse-reward settings. PPO's policy entropy starts at ~1.8 nats (MultiDiscrete) and decreases as the policy commits to navigation strategies; ppo_high_entropy (coef=0.05, +184.03) maintains higher entropy throughout, explaining its lower final mean relative to ppo_tight_clip (+364.73). REINFORCE's entropy decreases most smoothly over 1,000 episodes as the policy network specialises; reinforce_deep_net exhibits the most consistent entropy decline, correlated with its superior reward of +1,319.85. reinforce_no_baseline (+391.09) shows occasional entropy spikes from high-variance uncentred returns, but does not suffer full collapse over 1,000 episodes.

5.3 Generalisation

Generalisation was assessed by evaluating the best-performing models (dqn_large_buffer, ppo_tight_clip, reinforce_deep_net) on 10 additional episodes with randomised initial spacecraft positions, randomised orbital phases, and initial resources drawn uniformly from [0.5, 1.0]. DQN shows slight degradation: dqn_large_buffer averages −53.8 ± 19.4 on unseen starts vs −50.29 in training, reflecting the diverse replay buffer's broad state coverage. PPO generalises moderately: ppo_tight_clip averages +298.4 on unseen starts vs +364.73 in training (−18% degradation), suggesting partial overfitting to orbital configurations encountered during rollout collection. REINFORCE generalises best in absolute terms: reinforce_deep_net averages +1,104.7 on unseen starts vs +1,319.85 in training (−16% degradation), consistent with the policy network's capacity to encode robust heading and approach behaviours from a rich 26-dim observation. The dominant failure mode across all algorithms on unseen starts remains out-of-bounds (>90% of failed episodes), confirming trajectory planning from novel initial conditions as the core unsolved challenge.

6. Conclusion and Discussion

This project designed, implemented, and benchmarked three reinforcement learning algorithms on AstroLogic, a custom deep-space biosignature exploration environment with a rich 26-dimensional observation space, a 360-dimensional joint action space, and mixed dense-sparse rewards. The principal findings are summarised below.

REINFORCE achieves the best final performance across all 30 experiments, with reinforce_deep_net reaching +1,319.85 over 1,000 episodes. The key driver is network capacity: the [256, 128, 64] policy backbone provides sufficient representational power to encode the compositional navigation–detection–transmission task structure. The Monte Carlo return estimator, while high-variance, benefits from full 1,000-episode training in a way that faster on-policy methods cannot replicate at equivalent wall-clock cost.

PPO ranks second with a best final mean of +364.73 (ppo_tight_clip) over ~100–138 episodes. Its native MultiDiscrete support preserves compositional action structure, and conservative policy clipping (ε=0.15) prevents catastrophic forgetting of learned approach trajectories. The 100k-timestep budget limits how deeply PPO can exploit discovered rewards; extending to 500k timesteps would likely yield substantially higher final means.

DQN ranks last with a best final mean of −50.29 (dqn_large_buffer), never reaching positive territory. Action-space flattening (360 unordered discrete actions) destroys the compositional sub-action semantics that policy-gradient methods exploit naturally. The 300k replay buffer and PER are the

most effective hyperparameters, but cannot compensate for the fundamental structural mismatch between DQN's Q-function formulation and the MultiDiscrete action space.

Recommendations for future work:

- Curriculum learning: start with single-target, short-horizon episodes and progressively increase complexity to address the dominant failure mode (out-of-bounds navigation).
- Reward shaping: incorporate a Hohmann transfer efficiency bonus to guide agents toward energy-optimal inter-planetary trajectories.
- Hierarchical RL: decompose the task into a high-level goal selector (which body to visit next) and low-level navigation/instrument controllers.
- 3D visualisation: extend the current pygame 2D renderer to a Panda3D or Three.js scene for richer agent behaviour inspection and stakeholder demonstration.

AstroLogic demonstrates that even moderately complex custom RL environments expose significant algorithmic differences. The combination of dense shaping with sparse biosignature rewards, constrained resources, and a large compositional joint action space creates a benchmark that meaningfully differentiates DQN, PPO, and REINFORCE — providing a strong foundation for future research in autonomous space systems. The unexpected dominance of REINFORCE at the 1,000-episode scale — achieving +1,319.85 vs PPO's +364.73 and DQN's −50.29 — highlights the importance of training budget alignment when comparing RL algorithms: per-episode gradient updates give REINFORCE a data-efficiency advantage not captured by timestep-normalised comparisons.